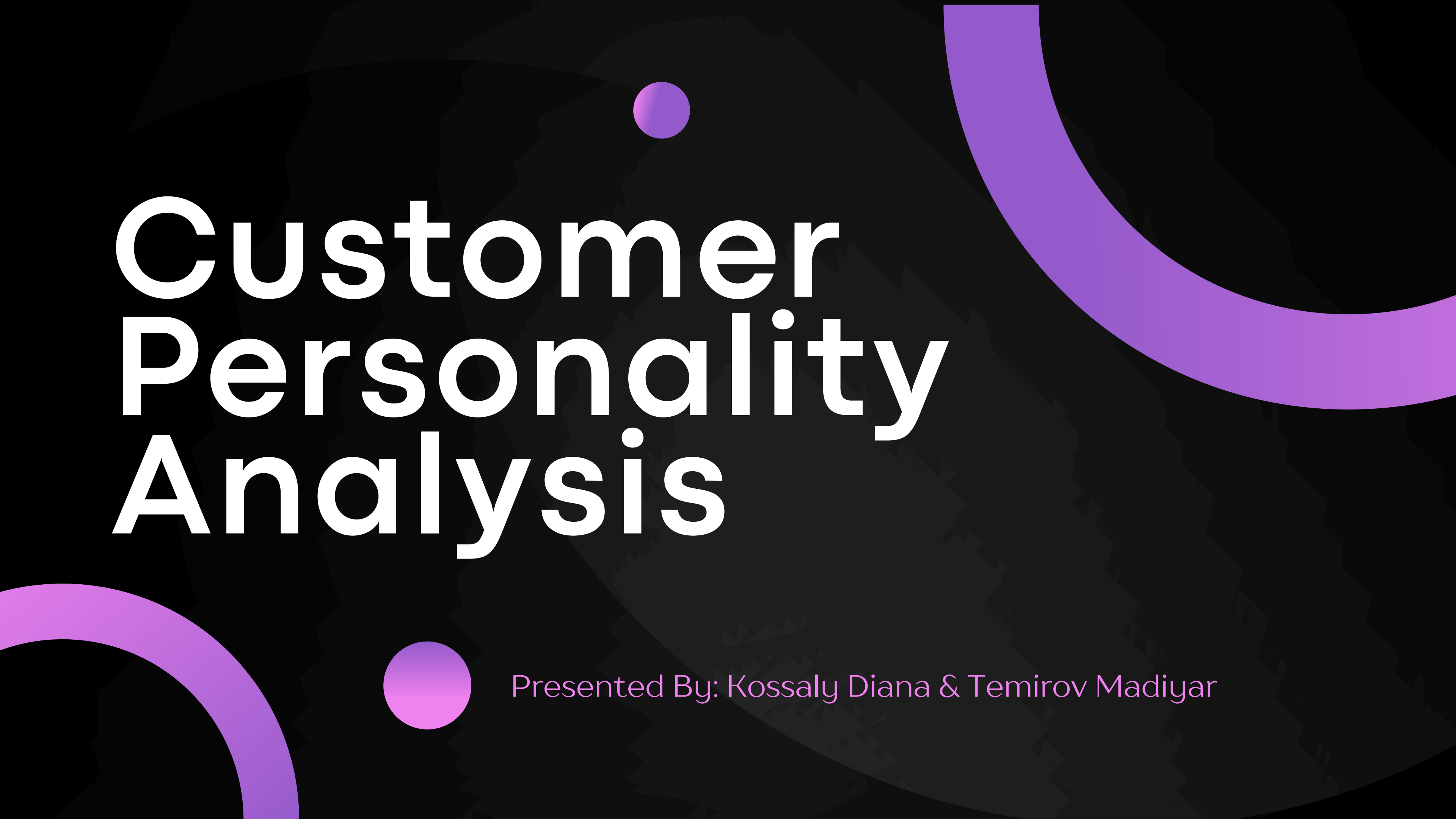


The background is black with several abstract purple shapes. There are two small solid purple circles, one near the top center and one near the bottom left. There are also two large, thick purple curved lines, one on the right side and one on the bottom left, resembling parts of large circles or arcs.

Customer Personality Analysis

Presented By: Kossaly Diana & Temirov Madiyar

Project Objective

Modern companies collect large amounts of customer data, but many businesses struggle to transform this data into useful marketing decisions.

Project Goal

Our goal was to build a complete AI marketing system that can:

- Analyze customer behavior
- Segment customers
- Predict purchasing behavior
- Personalize recommendations
- Analyze campaign performance
- Monitor customer sentiment
- Support ethical AI decision-making



Executive Marketing Overview

Dashboard Components

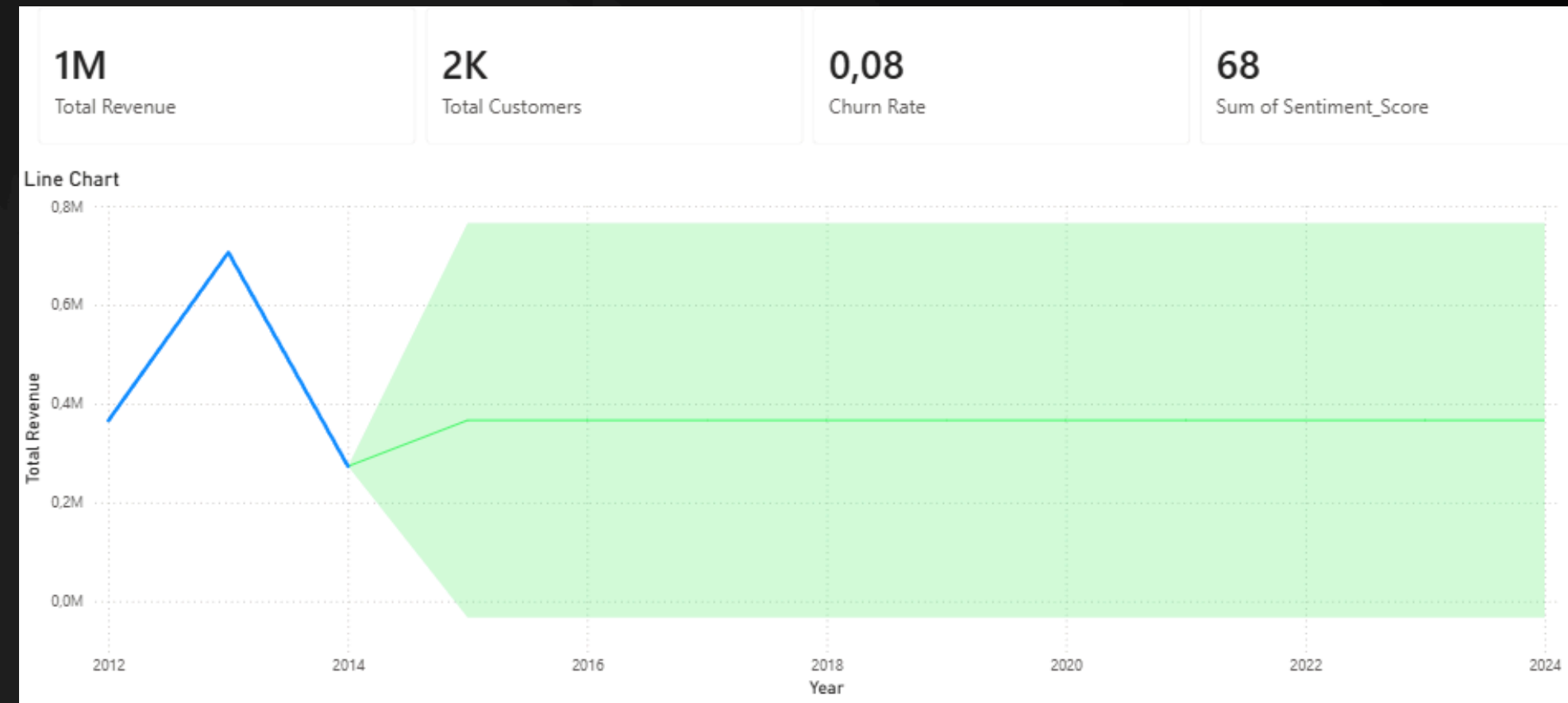
- Total Revenue
- Total Customers
- Churn Rate
- Average Sentiment Score

Forecasting

We used Power BI forecasting to predict future revenue trends for the next several months.

Brand Health Monitoring

A custom KPI visual was added to monitor overall brand performance.



Customer Segmentation Dashboard

AI Customer Segmentation

We applied K-Means clustering to group customers based on:

- Income
- Spending behavior
- Engagement level

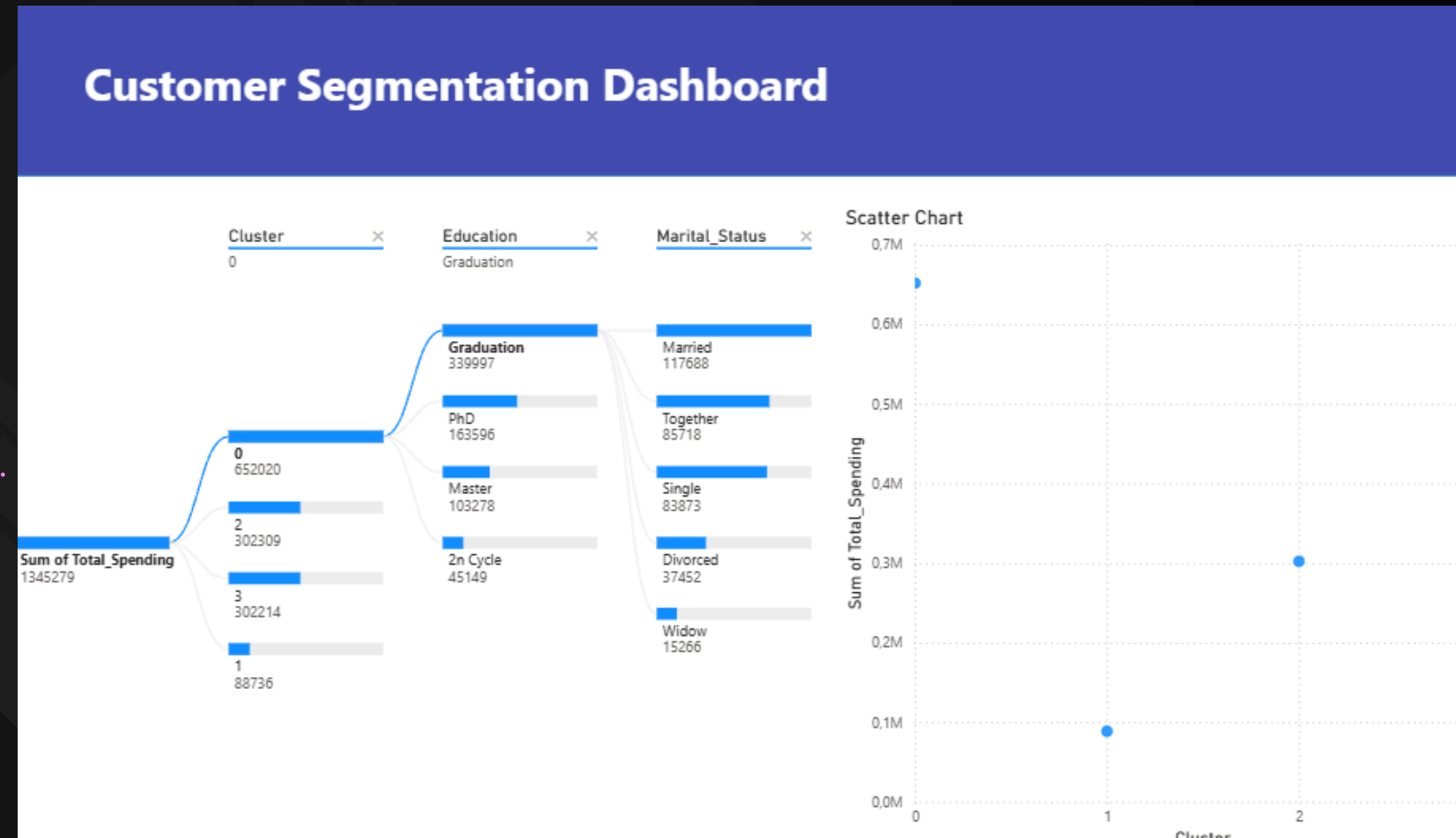
Scatter Plot Analysis:

The scatter plot visualizes customer clusters in real time.

Decomposition Tree:

The decomposition tree helps analyze:

- Which clusters spend the most
- Which education levels generate higher spending
- Which marital groups are most profitable



Predictive Analytics Dashboard

We analyzed:

- Churn probability
- Campaign response likelihood
- Customer behavior trends

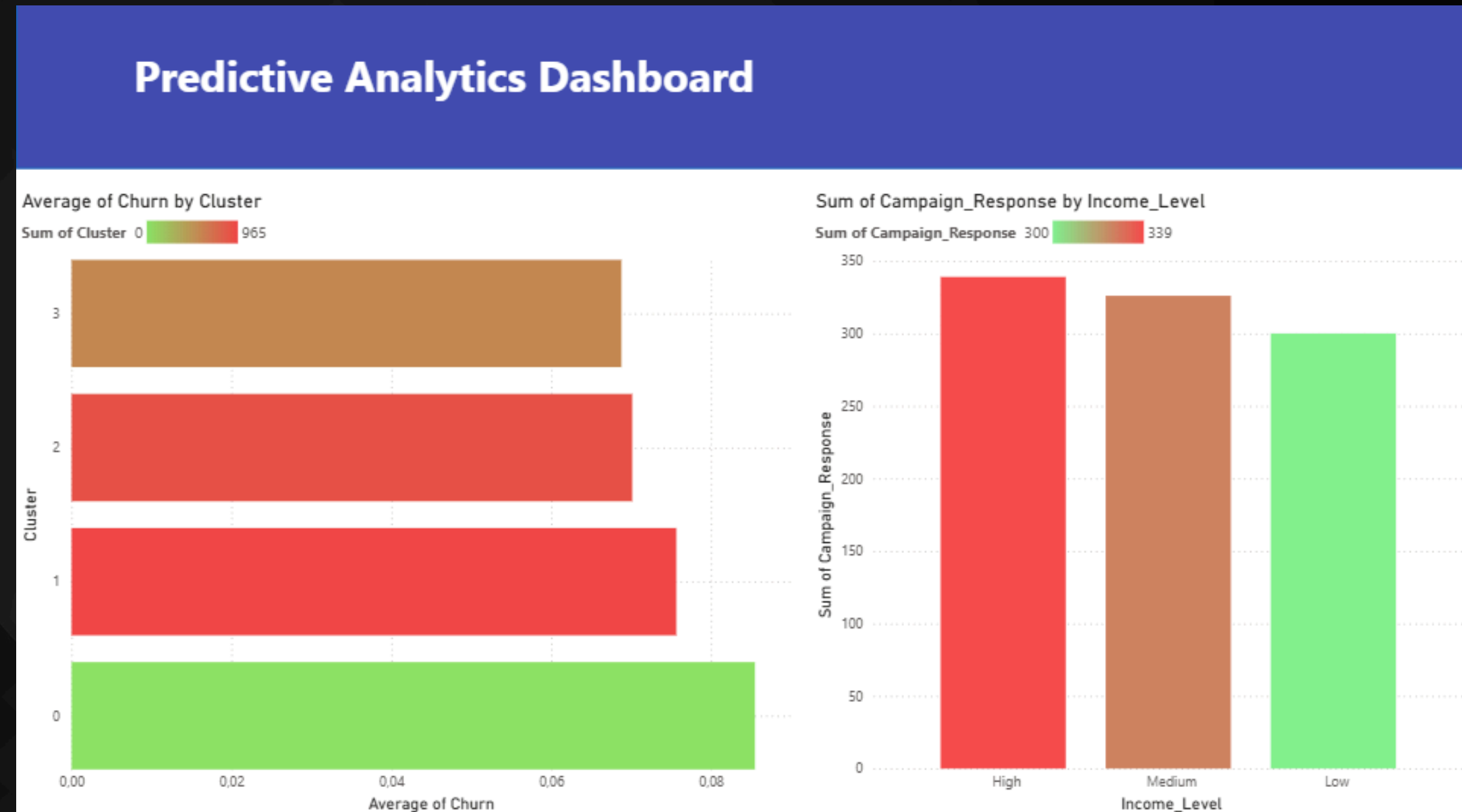
Visual Features

- Conditional formatting highlights high-risk churn groups
- Income-level analysis predicts campaign success probability

Business Value

Predictive analytics helps businesses:

- Reduce customer churn
- Improve campaign targeting
- Optimize marketing budgets



Personalization & Recommendation Dashboard

We analyzed:

- Customer loyalty
- Preferred product categories
- Engagement behavior
- Preferred sales channels

Recommendation Insights

The matrix visual helps compare spending and engagement across customer groups.

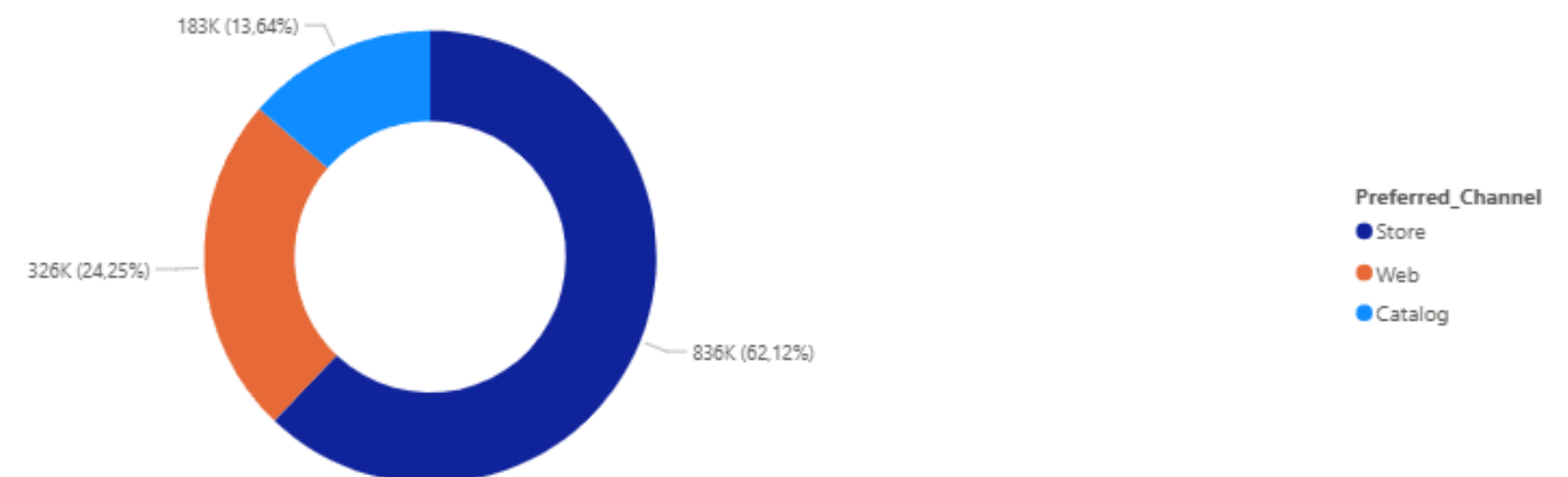
Channel Performance

The donut chart shows which sales channels generate the highest customer spending.

Personalization & Recommendation Dashboard

Cluster	Fish	Fruits	GoldProds	Meat	Sweet	Wines	Total
1	4541	1081	14246	16020	807	52041	88736
3				59196		243018	302214
2	577		1726	23597	475	275934	302309
0	2099		2213	193915	277	453516	652020
Total	7217	1081	18185	292728	1559	1024509	1345279

Donut Chart



Campaign Performance Dashboard

We compared:

- AcceptedCmp1
- AcceptedCmp2
- AcceptedCmp3
- AcceptedCmp4
- AcceptedCmp5
- Final Campaign Response

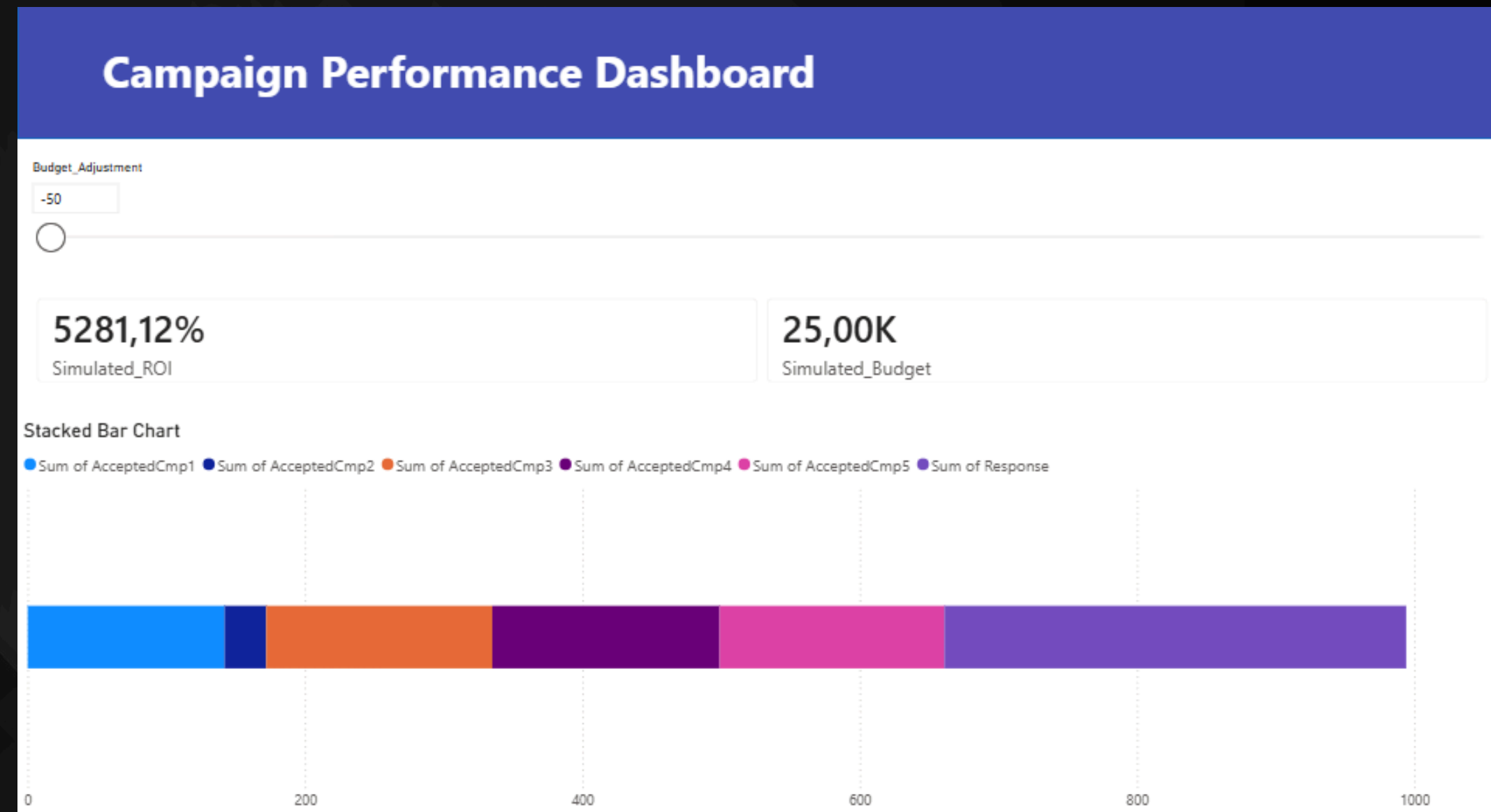
ROI Scenario Modeling

We created a What-if Parameter called Budget Adjustment.

Business Impact

This allows marketers to simulate:

“What happens to ROI if marketing budgets increase or decrease?”



Sentiment & Social Media Dashboard

We analyzed customer reviews using Natural Language Processing techniques.

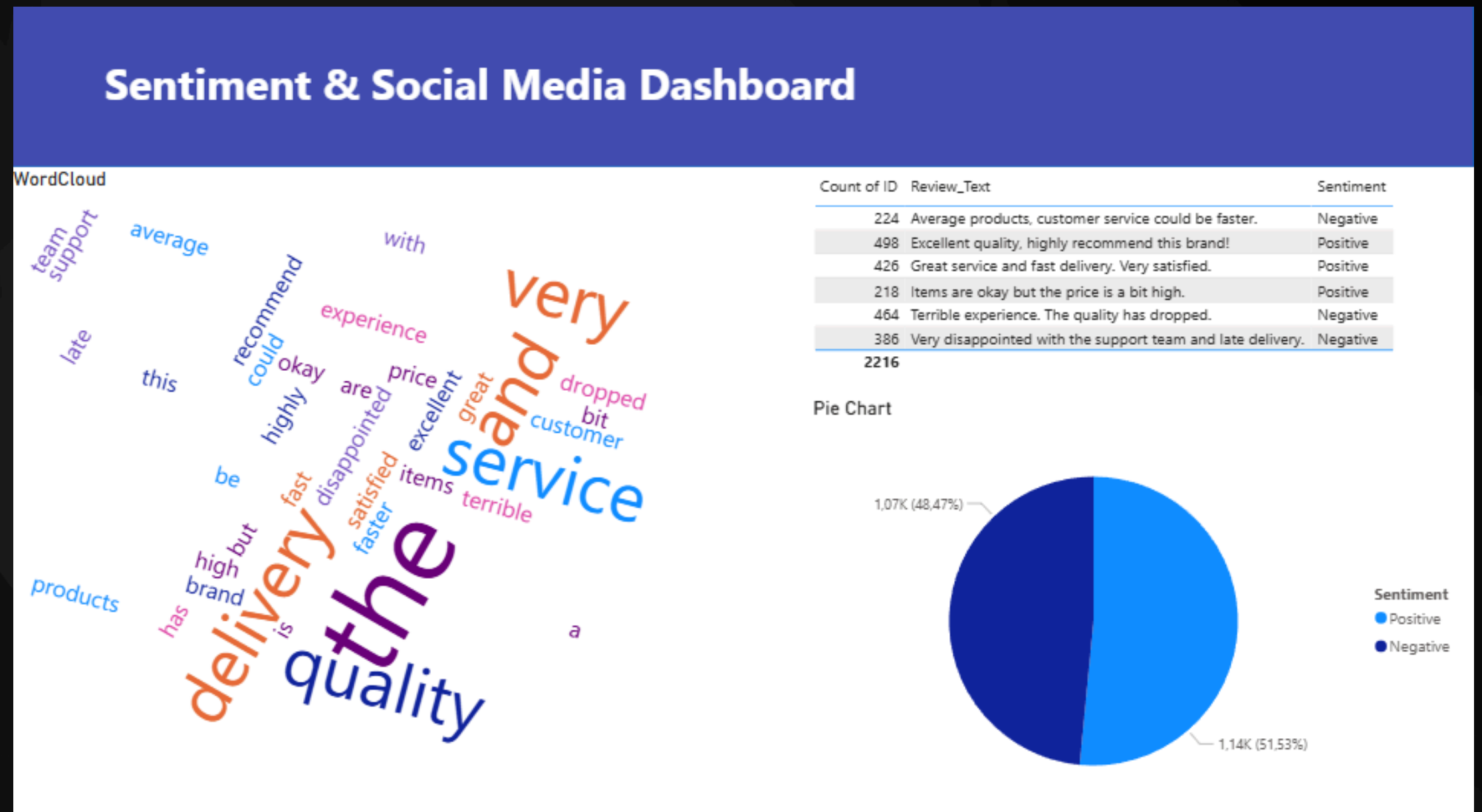
Dashboard Features

- Word Cloud visualization
- Sentiment distribution
- Negative and Positive reviews filtering

Key Insights

Customer sentiment provides valuable feedback about:

- Customer satisfaction
- Product perception
- Brand reputation



Ethics, Privacy & Responsible AI

We evaluated:

- Bias risks
- Customer privacy
- Ethical AI usage
- Data transparency

Bias Monitoring

We analyzed churn behavior across:

- Age groups
- Marital status
- Demographic categories

Privacy Checklist

- Data anonymization
- Customer consent
- Opt-out mechanisms

Ethics, Privacy & Risk Dashboard

Privacy Checklist Items

- ☐ Customer Opt-out Compliance
- ☐ Data Anonymization (ID Hashing)
- ☐ GDPR & Fair Treatment Verified

AI Privacy Statement: All customer PII (Personally Identifiable Information) in this dashboard has been fully anonymized. The predictive models exclude sensitive features to prevent algorithmic bias.

Marital_Status	Sum of Churn	Sum of Age
Absurd	0	98
Alone	0	153
Divorced	16	13392
Married	76	46648
Single	34	24694
Together	38	32233
Widow	6	4958
YOLO	0	102
Total	170	122278

AI Marketing Strategy & Recommendations

Strategic Recommendations

1. Move more budget to Campaign 4 and 5 because they performed better than Campaign 2.
2. Create special loyalty offers for high-value customers to reduce churn.
3. Focus more on Web and Store channels because customers showed the highest satisfaction there.
4. Use recommendation systems to suggest related products and increase sales.
5. Send personalized email offers to customers with high churn risk.

Conclusion

Our AI system helps businesses make smarter marketing decisions using customer data, machine learning, and Power BI.



THANK YOU